

Assignment 2 — Geostrophic transport and relating AMOC series

Course: Data Analysis for Physical Oceanography (MSc)
Weight: Assignment 2 of 2
Issued: 6 July 2026
Due: 4 September 2026 (submitted as a git repository)
Effort: ~3 weeks

Aim

Two things, in two parts. First, **compute** an ocean transport yourself from hydrography using geostrophy (thermal wind), and check it against a published product. Second, **relate time series to one another** — quantify how a single series remembers its past, whether it trends, and whether two series lead, lag, or move together.

The two parts are deliberately independent: **Part 2 uses time series we provide, so you can complete it in full even if your Part 1 geostrophy calculation does not reproduce the published transport well.** A geostrophic estimate that only half works is still a good result to analyse and explain.

Learning outcomes

By the end you will be able to:

1. Compute a geostrophic (thermal-wind) transport from boundary temperature and salinity profiles, and state the choices (reference level, integration depth) that it depends on.
2. Estimate the persistence of a series (autocorrelation, decorrelation timescale) and turn it into an **effective sample size**.
3. Fit a trend and judge its significance **honestly**, accounting for autocorrelation.
4. Put two series on a comparable footing (common time grid, **matched filtering**) and cross-correlate to find a lead/lag after removing the seasonal cycle.
5. Distinguish correlation strength (r , r^2) from statistical significance, and say when an apparent relationship is only suggestive.
6. Package the analysis as tested code in a reproducible git repository.

The datasets

All via the **AMOCatlas** package (standardised names on a **TIME** coordinate):

```
from amocatlas import read
```

```
# 26N transports: MOC, TRANS_FC, TRANS_EKMAN, TRANS_UMO
transports = read.rapid()
```

```
# 26N boundary hydrography for Part 1 (TEMP_/PSAL_ WEST & EAST on PRESSURE)
ts = read.rapid(file_list=["ts_gridded.nc"], transport_only=False)
```

```
# 26N overturning in density coordinates: MOC_SIGMA0
mt = read.rapid(file_list=["meridional_transports.nc"], transport_only=False)
```

```
# 47N NOAC overturning (monthly): MOC_SIGMA0
noac = read.read_47n()
```

Loaders for each are provided in `correlation_trends.data_io`. Report the length, sampling, and time span of anything you use — that is part of the task.

Required work

Part 1 — Geostrophic transport (specified series)

From the RAPID 26°N boundary hydrography (`ts_gridded`), compute the **upper mid-ocean geostrophic transport** and compare it with RAPID’s published `TRANS_UMO`.

1. Form the dynamic height at the western and eastern boundaries (TEOS-10; helpers provided), difference them for the zonally-integrated geostrophic transport per unit depth (thermal wind), and **integrate over depth to the depth of the AMOC maximum** (~1100 m) for a transport time series in Sv. Explain why the *upper* mid-ocean transport is not a full-depth integral.
2. Compare with `TRANS_UMO`: **overlay** the two series, report the **correlation**, and make a **scatter** plot with the regression line. Comment on the mean offset and the extra variance in your estimate, and why it differs from the published product (which also splits the interior at the Mid-Atlantic Ridge and applies a mass-balance adjustment).

This part has a known target, so we can check your answer against it.

Part 2 — Evaluating time series

2A — Single series: seasonal cycle and trend (use MOC and `TRANS_UMO`).

For **both** the 26°N MOC and `TRANS_UMO`:

1. Compute and remove the **seasonal cycle** — show the monthly climatology, and plot the **deseasonalised series with the overall mean kept** (draw the deseasonalised line darker and thicker over the raw series).
2. Show the **autocorrelation** with the **decorrelation timescale** T^* marked (a line at T^* is enough; shading optional); T^* sets the effective sample size used below.
3. Fit a **linear trend** and test whether it differs from zero. Report the slope with its standard error, the slope in units of that error (σ from zero), and the p-value **using the effective sample size** (from the decorrelation timescale), not the raw one. State whether each trend is significant at 95%, and why.

These are series we specify, so there is a definite answer to grade against.

2B — A pair of series: cross-correlation and lead/lag (your choice of pair).

Pick **any two** series — e.g. 26°N vs 47°N `MOC_SIGMA0`; `TRANS_EKMAN` vs MOC; `TRANS_UMO` vs `TRANS_FC`; or your Part 1 transport vs `TRANS_UMO`.

1. **Match the processing first.** If the two series are filtered differently (e.g. one is a 10-day low-pass, the other is not), **pre-filter the less-filtered series to match** before comparing — otherwise the extra variance in one series biases the correlation. Put both on a common time grid. (*A quick before/after figure of the matched series is good “show your working”.*)

2. Remove the seasonal cycle from each, then **cross-correlate** and identify any **lead/lag** (which leads, by how long — state your sign convention). Show the cross-correlation **raw and deseasonalised**, and make a **scatter at the peak lag**.
3. Interpret honestly: given the record length and the persistence of the series, how many independent samples do you really have — is the relationship significant, or only suggestive?

Part 3 — One calculation of your choice

Add **one** further analysis, to show some initiative. Some options:

- the depth **sensitivity** of your Part 1 transport (integrate to 700 / 1000 / 1100 m and compare the mean and correlation);
- a **second pair** of series, or the same pair on a different array (OSNAP, MOVE);
- **confidence intervals** on a trend or correlation via a resampling method;
- **biased vs unbiased** autocorrelation/cross-correlation, and where they diverge;
- your own idea — state the question and answer it.

What to implement

Work in the `correlation_trends` package (the Lecture 4 starter). **Four** functions are left as stubs for you to implement; the `pytest` checks encode their contract:

- `correlation.effective_dof` — effective sample size $\text{EDOF} = \text{record}/(2T^*)$ from the (provided) integral timescale.
- `correlation.cross_correlation` — normalised cross-correlation with a lag axis.
- `seasonal.remove_seasonal_cycle` — subtract the monthly climatology **and add the overall mean back**, so the deseasonalised series keeps its level.
- `trends.trend_with_significance` — slope, standard error, slope/ σ , and the autocorrelation-aware p-value (uses your `effective_dof`).

Provided (do not re-implement): `autocorr`, `integral_timescale`, `seasonal_climatology`, the geostrophy helpers (`to_teos10`, `dynamic_height`), and the `data_io` loaders. Keep functions small and tested; your figure script should import them, not re-implement them.

Required tests

Your repository must include (at least) two passing `pytest` tests:

1. **Cross-correlation recovers a known lag** — for a series and a shifted copy, the peak lands at the expected lag.
2. **Trend significance behaves** — the slope of a planted trend is recovered, and for an autocorrelated series the effective standard error is larger than the naive one (so the honest p-value is larger).

Deliverables

A **git repository** containing:

1. The code: your implemented `effective_dof`, `cross_correlation`, `remove_seasonal_cycle`, and `trend_with_significance`, plus the script/notebook that reproduces your figures from the raw download.
2. A passing test suite (`pytest` runs green), including the two required tests.

3. The figures (see *Figures you'll likely generate*, below).
4. A short writeup ($\sim 2\text{--}3$ pages, `.rst` or `.md`) answering Parts 1 and 2 in prose, with honest interpretation of significance.

Submission & git workflow

Submit the URL of your repository and your report on the Moodle. Work on a branch, open a pull request into `main`, and **tag** the commit you want marked (`git tag assignment2 && git push -tags`). Tests must run from a clean checkout (include a `requirements.txt`).

Assessment (rubric — see separate code rubric)

Tips & common pitfalls

- **Match the filtering before you compare.** Correlating a 10-day-low-passed series against an unfiltered one is not a fair comparison — pre-filter the less-filtered one to the same window first, and put both on a common time grid.
- **Deseasonalise before you cross-correlate.** Two series that share an annual cycle correlate at ± 12 months even when otherwise unrelated.
- **Significance is not the correlation.** A large r with few independent samples can be insignificant; report the effective sample size.
- **Mind the lead/lag sign** — decide and state which series leading means a positive lag, and sanity-check it on a synthetic shifted series.
- **Geostrophy depends on your choices** — reference level and integration depth change the answer; show the sensitivity rather than hiding it.

Figures you'll likely generate

A rough guide, not a checklist — whatever tells the story clearly:

- **Part 1:** geostrophic transport vs `TRANS_UMO` — an overlay and a scatter.
- **Part 2A** (for `MOC` and `TRANS_UMO`): the seasonal cycle (monthly climatology); the deseasonalised series (mean kept) over the raw; the autocorrelation with the decorrelation timescale marked; and the trend fit with its honest confidence band.
- **Part 2B:** the cross-correlation vs lag (raw and deseasonalised); and the scatter at the peak lag.
- **Optional (show your working):** the two series after matched filtering, before cross-correlating; biased vs unbiased autocorrelation.

Resources

- Lecture 4 deck and starter code (`correlation_trends`, `make_figures.py`).
- Emery & Thomson, *Data Analysis Methods in Physical Oceanography*, Ch. 5 (§5.3 *Correlation and covariance*; §5.5 *Regression*; §5.8 *Cross-spectral analysis*).
- McCarthy et al. (2015), *Measuring the Atlantic Meridional Overturning Circulation at 26°N* (upper mid-ocean transport; level of no motion).
- Wett et al. (2023), *Meridional connectivity of a 25-year observational AMOC record at 47°N*, GRL.
- AMOCatlas: <https://github.com/AMOCcommunity/AMOCatlas>